

Bayesian Optimization with Gradients

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MIPT, 2025

March 25, 2025

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Definitions

What is Bayesian Optimization?

Bayesian Optimization (BO) is a **sequential** global optimization strategy for expensive **black-box** functions $f : \mathbb{A} \rightarrow \mathbb{R}$, where $\mathbb{A} \subset \mathbb{R}^d$, and:

- f has 1) no closed form, 2) noisy evaluations, or 3) costly gradients.
- Goal: Find $\mathbf{x}^* = \arg \min_{\mathbf{x} \in \mathbb{A}} f(\mathbf{x})$ with few function evaluations.

Key Steps of Bayesian Optimization

- 1 Use the a special function to decide at which input $\mathbf{x} \in \mathbb{A}$ to query f next (use a so-called **acquisition function**);
- 2 Evaluate $f(\mathbf{x})$;
- 3 **Update the posterior** based on the new data $\mathbf{x}$ and $f(\mathbf{x})$.

Motivation

- Bayesian optimization excels in globally optimizing expensive-to-evaluate multimodal objective functions. However most methods **does not rely on derivative information**.
- **The main idea** is to propose a method using derivative information to reduce the calculation of the function.
- This paper introduces a novel Bayesian optimization algorithm, **the derivative-enabled knowledge gradient (d-KG)**, which is one-step Bayes-optimal, asymptotically consistent, and provides greater one-step value of information than in the derivative-free setting.

Proposed Method

Assumption

- To model the unknown function f , a Gaussian Process (GP) prior is used. A GP is specified by its mean function $\mu : \mathbb{A} \rightarrow \mathbb{R}$ and covariance function $K : \mathbb{A} \times \mathbb{A} \rightarrow \mathbb{R}$. $\mathbb{A} \subset \mathbb{R}^d$ is the domain.
- Observations include noisy function values $y(\mathbf{x})$ and gradients $\nabla y(\mathbf{x})$.

Idea

Since the gradient is a linear operator, the gradient of a GP is also a GP. To leverage derivative information, the joint distribution of $f(\mathbf{x})$ and its gradient $\nabla f(\mathbf{x})$ is modeled as a multi-output GP.

Proposed Method

Gaussian Process with Derivatives

The joint distribution of $f(\mathbf{x})$ and its gradient $\nabla f(\mathbf{x})$ is modeled as a multi-output GP:

$$\tilde{\mu}(\mathbf{x}) = (\mu(\mathbf{x}), \nabla \mu(\mathbf{x}))^T$$

$$\tilde{K}(\mathbf{x}, \mathbf{x}') = \begin{pmatrix} K(\mathbf{x}, \mathbf{x}') & J(\mathbf{x}, \mathbf{x}') \\ J(\mathbf{x}', \mathbf{x})^T & H(\mathbf{x}, \mathbf{x}') \end{pmatrix}$$

where $J(\mathbf{x}, \mathbf{x}')$ is the Jacobian and $H(\mathbf{x}, \mathbf{x}')$ is the Hessian of $K(\mathbf{x}, \mathbf{x}')$.

Distribution of Observations

The joint distribution of observations can be written as:

$$(y(\mathbf{x}), \nabla y(\mathbf{x}))^T \mid f(\mathbf{x}), \nabla f(\mathbf{x}) \sim \mathcal{N} \left((f(\mathbf{x}), \nabla f(\mathbf{x}))^T, \text{diag}(\sigma^2(\mathbf{x})) \right)$$

where $\sigma^2(\mathbf{x})$ is the variance of the observation noise.

Proposed Method

Posterior Distribution in GP

The posterior distribution in Gaussian Processes (GP) is described by an updated mean function $\tilde{\mu}^{(n)}(\mathbf{x})$ and covariance function $\tilde{K}^{(n)}(\mathbf{x}, \mathbf{x}')$. These functions are computed based on observations $(y, \nabla y)^{(1:n)}$ at points $\mathbf{X}$.

$$\begin{aligned}\tilde{\mu}^{(n)}(\mathbf{x}) &= \tilde{\mu}(\mathbf{x}) + \tilde{K}(\mathbf{x}, \mathbf{X}) \cdot \left(\tilde{K}(\mathbf{X}, \mathbf{X}) + \text{diag} \left\{ \sigma^2(\mathbf{x}^{(i)}) \right\} \right)^{-1} \left((y, \nabla y)^{(1:n)} - \tilde{\mu}(\mathbf{X}) \right) \\ \tilde{K}^{(n)}(\mathbf{x}, \mathbf{x}') &= \tilde{K}(\mathbf{x}, \mathbf{x}') - \tilde{K}(\mathbf{x}, \mathbf{X}) \left(\tilde{K}(\mathbf{X}, \mathbf{X}) + \text{diag} \left\{ \sigma^2(\mathbf{x}^{(i)}) \right\} \right)^{-1} \tilde{K}(\mathbf{X}, \mathbf{x}')\end{aligned}$$

Proposed Method

Acquisition Function

The d-KG acquisition function is defined as:

$$d\text{-KG} \left(\mathbf{z}^{(1:q)} \right) = \min_{\mathbf{x} \in \mathbb{A}} \tilde{\mu}_1^{(n)}(\mathbf{x}) - \mathbb{E}_n \left[\min_{\mathbf{x} \in \mathbb{A}} \tilde{\mu}_1^{(n+q)}(\mathbf{x}) \mid \mathbf{x}^{(n+1):(n+q)} = \mathbf{z}^{(1:q)} \right]$$

where $\tilde{\mu}_1^{(n)}(\mathbf{x})$ is the expected value of $f(\mathbf{x})$ under the posterior distribution after n evaluations.

Efficient Computation

The algorithm uses a novel discretization-free method to compute the acquisition function, enhancing speed and accuracy. It supports batch evaluations and handles noisy, incomplete derivative information. An unbiased gradient estimator of d-KG is used within stochastic gradient ascent to maximize d-KG.

d -KG Acquisition Function and Algorithm

We define the d -KG acquisition function for observing only the function value and the derivative with direction θ at $\mathbf{z}^{(1:q)}$ as:

$$d\text{-KG}(\mathbf{z}^{(1:q)}, \theta) = \min_{\mathbf{x} \in \mathbb{A}} \tilde{\mu}_1^{(n)}(\mathbf{x}) - \mathbb{E}_n \left[\min_{\mathbf{x} \in \mathbb{A}} \tilde{\mu}_1^{(n+q)}(\mathbf{x}) \mid \mathbf{x}^{(n+1):(n+q)} = \mathbf{z}^{(1:q)}; \theta \right]$$

where conditioning on θ is here understood to mean that $\tilde{\mu}_1^{(n+q)}(\mathbf{x})$ is the conditional mean of $f(\mathbf{x})$ given $y(\mathbf{z}^{(1:q)})$ and $\theta^T \nabla y(\mathbf{z}^{(1:q)})$.

Algorithm 1 d -KG with Relevant Directional Derivative Detection

- 1: **for** $t = 1$ to N **do**
 - 2: $(\mathbf{z}^{(1:q)*}, \theta^*) = \arg \max_{\mathbf{z}^{(1:q)}, \theta} d\text{-KG}(\mathbf{z}^{(1:q)}, \theta)$
 - 3: Augment data with $y(\mathbf{z}^{(1:q)*})$ and $\theta^{*T} \nabla y(\mathbf{z}^{(1:q)*})$.
 - 4: Update our posterior on $(f(\mathbf{x}), \nabla f(\mathbf{x}))$.
 - 5: **end for**
 - 6: **return** $\mathbf{x}^* = \arg \min_{\mathbf{x} \in \mathbb{A}} \tilde{\mu}_1^{Nq}(\mathbf{x})$
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Including all d partial derivatives can be computationally prohibitive since GP inference scales as $\mathcal{O}(n^3(d+1)^3)$

Experiments: Synthetic Test Functions

The algorithm was tested on various synthetic benchmarks, including Branin, Rosenbrock, Ackley, Hartmann, Levy, and Cosine mixture functions.

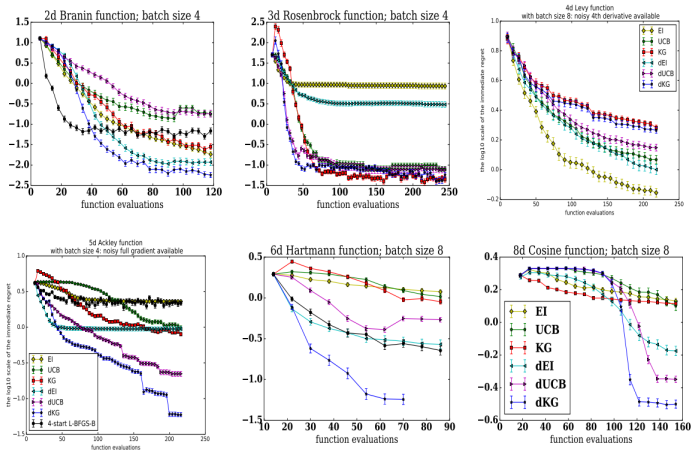


Figure 2: The average performance of 100 replications (the log10 of the immediate regret vs. the number of function evaluations). *d-KG* performs significantly better than its competitors for all benchmarks except Levy function. In Branin and Hartmann, we also plot black lines, which is the performance of BFGS.

Experiments: Real-World Applications

The method was applied to real-world problems such as hyperparameter tuning for weighted k-nearest neighbors, logistic regression, deep learning (tune logistic regression and a feedforward neural network with 2 hidden layers on the MNIST dataset), and kernel learning.

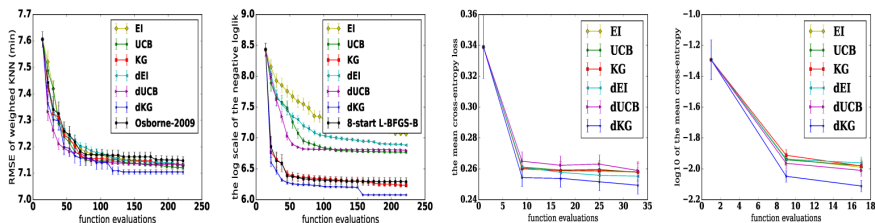


Figure 3: Results for the weighted KNN benchmark, the spectral mixture kernel benchmark, logistic regression and deep neural network (from left to right), all with batch size 8 and averaged over 20 replications.

- 1 **Main article:** Wu J. et al. Bayesian Optimization with Gradients // NIPS 2017.